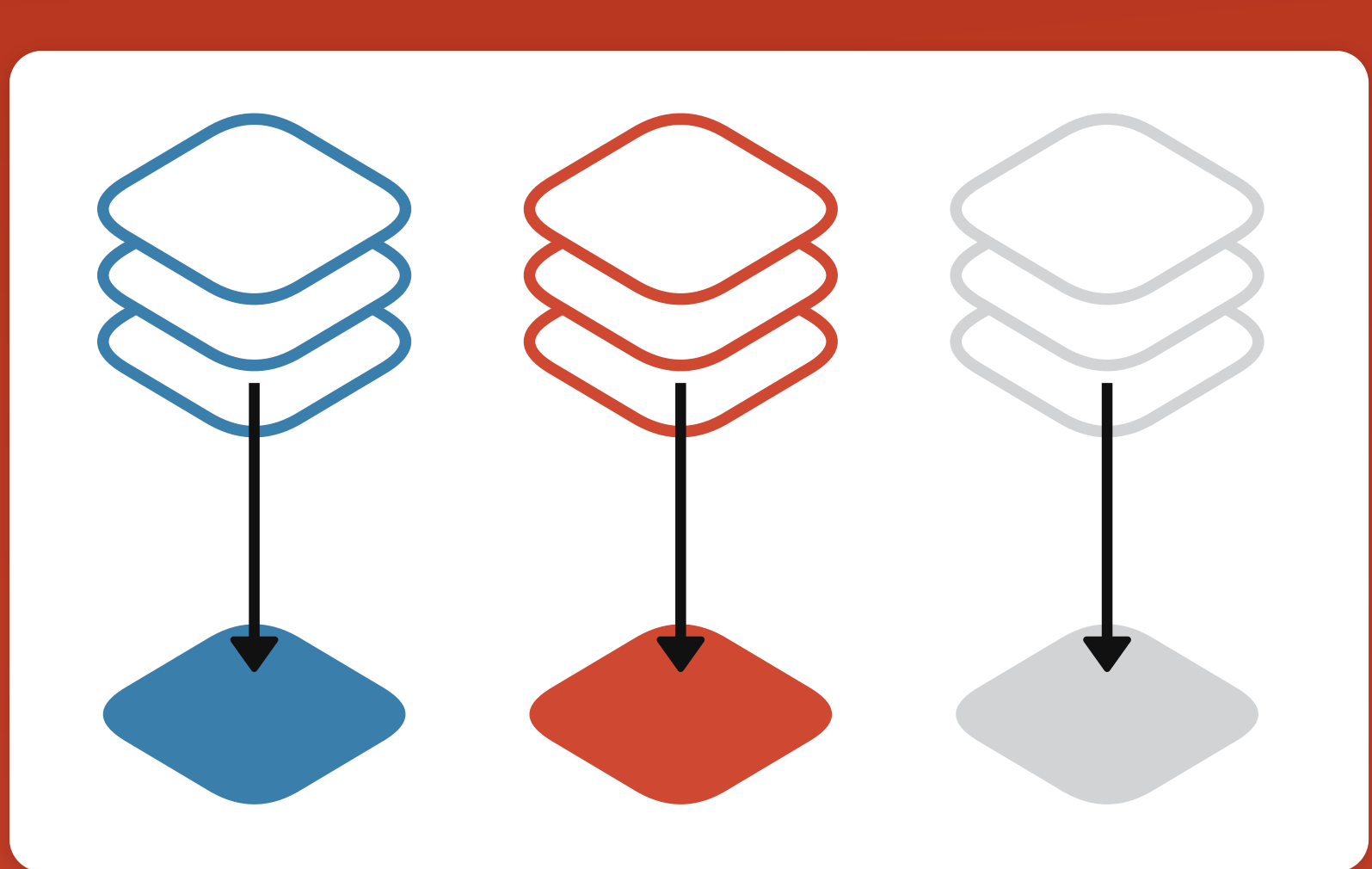
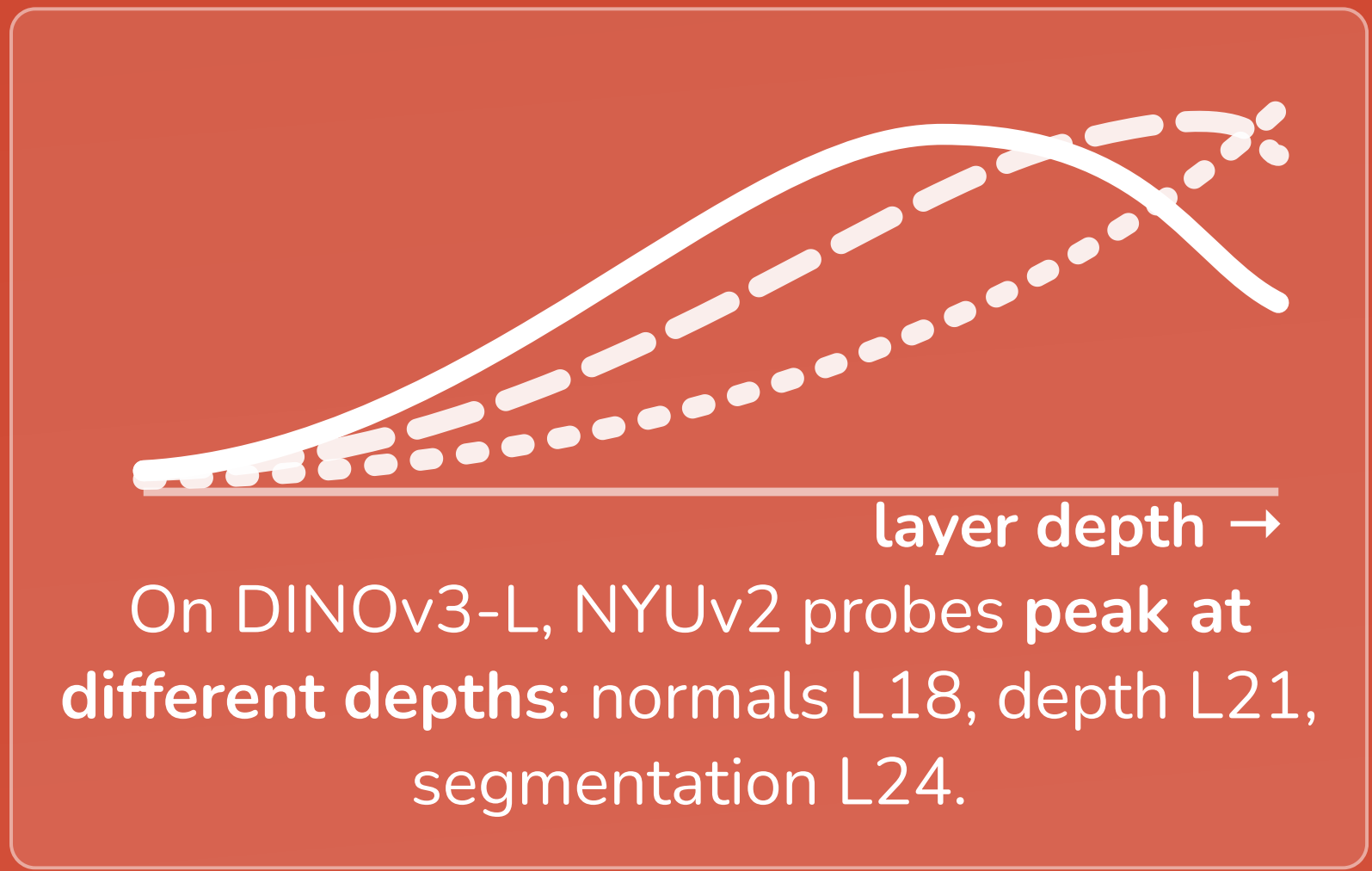




Which layer of a frozen vision foundation model should you probe?

None alone.

For dense prediction, try **AMDF**: it fuses **all** layers, for **every** patch.



1.48
Best mean rank over 5 benchmarks (4 dense + camera pose) and 7 frozen backbones.

≈1/4
of the **trainable parameters** of concatenating all layers (1.6M vs 6.8M, DINOv3-L, normals).

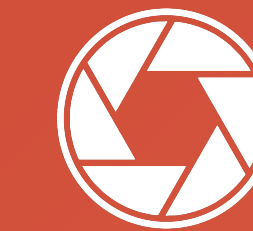
Frozen backbone · no layer search · one probe, any head

AMDF: Holistic Probing of Vision Foundation Models with Adaptive Multi-Layer Dense Fusion

Turhan Can Kargin^{1,2} · Bill Psomas^{3,4} · Paul Wagner⁵ · Giorgos Tolias^{3,4} · Bartosz Zieliński^{1,2} · Marcin Przewięźlikowski^{1,2}

¹ Jagiellonian University in Kraków ² GMUM ³ Czech Technical University in Prague ⁴ VRG ⁵ RWTH Aachen University

Contact: turhancan.kargin@doctoral.uj.edu.pl



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Different tasks live at different depths

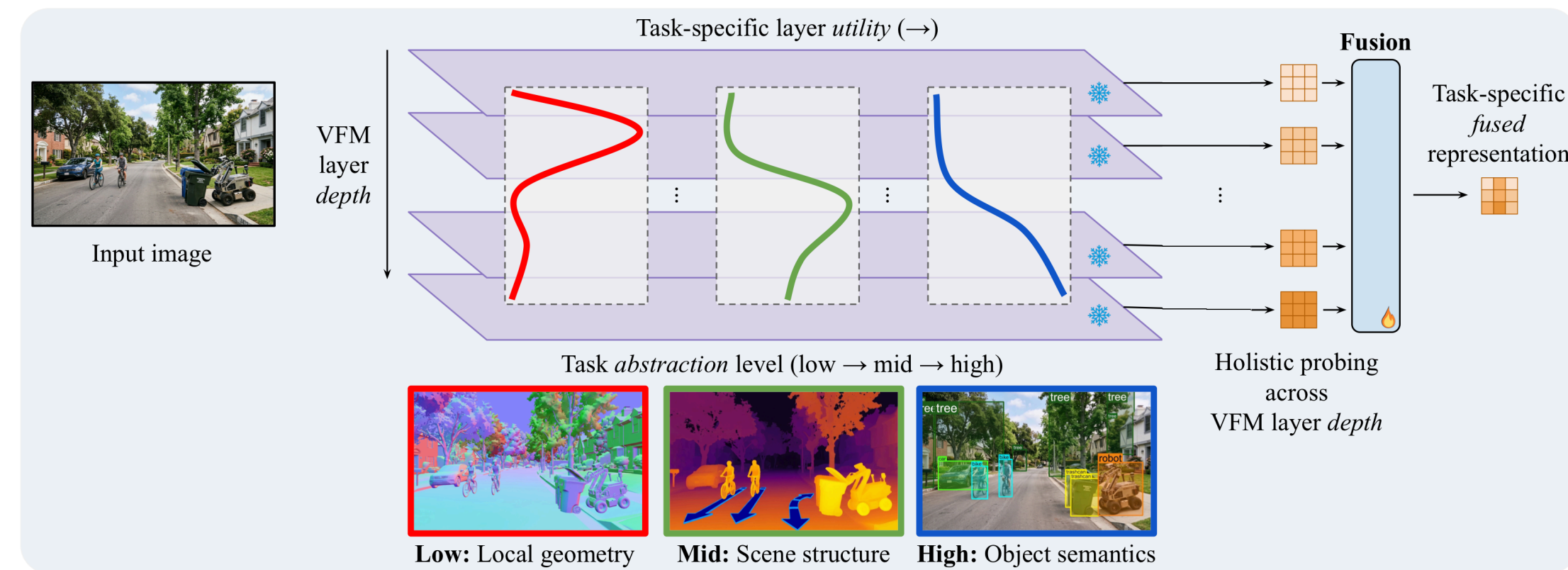


Fig. 1: Geometry peaks in middle layers, semantics near the end. AMDF reads every frozen layer and fuses them per patch.

Problem #1: the last layer is not the best

✗ Probing only the final layer misses what a task needs.

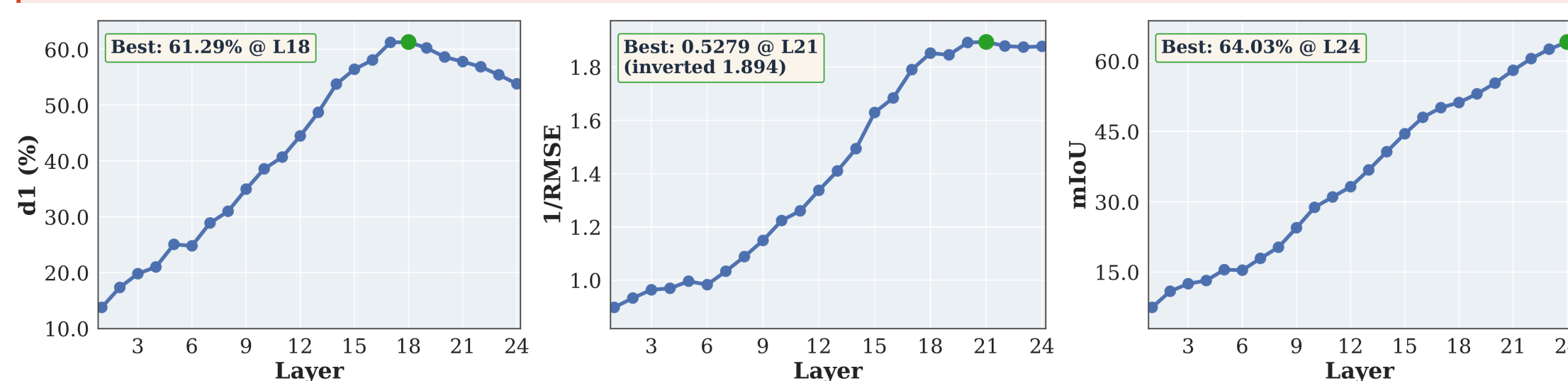


Fig. 2: One probe per layer of a frozen DINOv3-L on NYUv2: the best layer moves deeper as the task gets more semantic (L18 → L21 → L24).

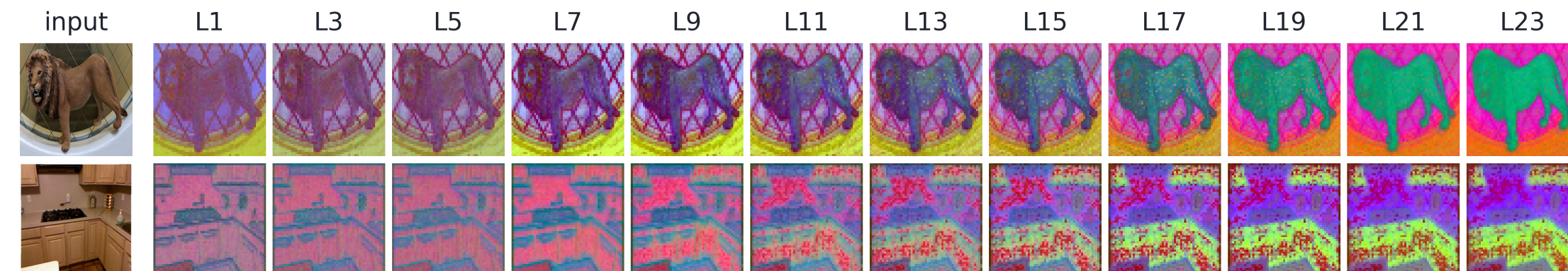


Fig. 3: Layers see differently: frozen PESpatial-L patch features (one PCA per image, 770 px input); object-background separation appears only in deep layers.

Problem #2: using all layers is costly

✗ Concatenating all layers works, but needs **6.8M** trainable parameters (vs **1.6M** for AMDF, DINOv3-L, incl. head) — and still treats every patch the same.

✓ **AMDF** learns, for every patch, how much of each layer to use.

Setup

7 frozen backbones

DINOv3-S/B/L · MAE-L · SigLIP2-L · V-JEPA2.1-L · PESpatial-L

6 benchmarks

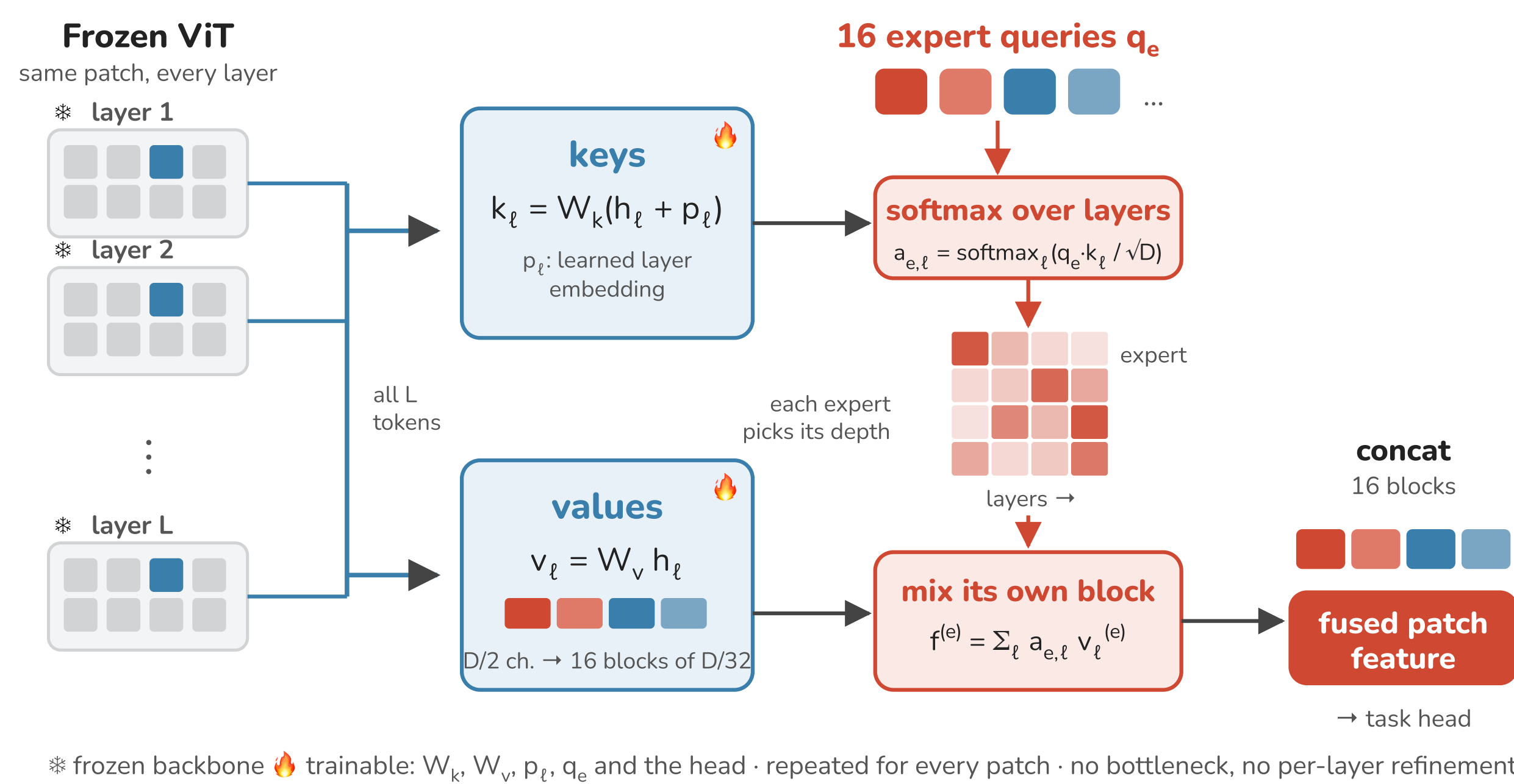
NYUv2 normals, depth, segmentation · Taskonomy · NAVI camera pose · CUB-200

Fair comparison

same recipe for every row of a table · mean of up to 3 seeds (a few cells have 1–2) · best-epoch validation metric

Solution: AMDF — depth-aware fusion

For every patch, 16 learned expert queries each take a **softmax over all layers** and mix their own block of channels, so each expert picks its own depth.



One probe, three kinds of heads

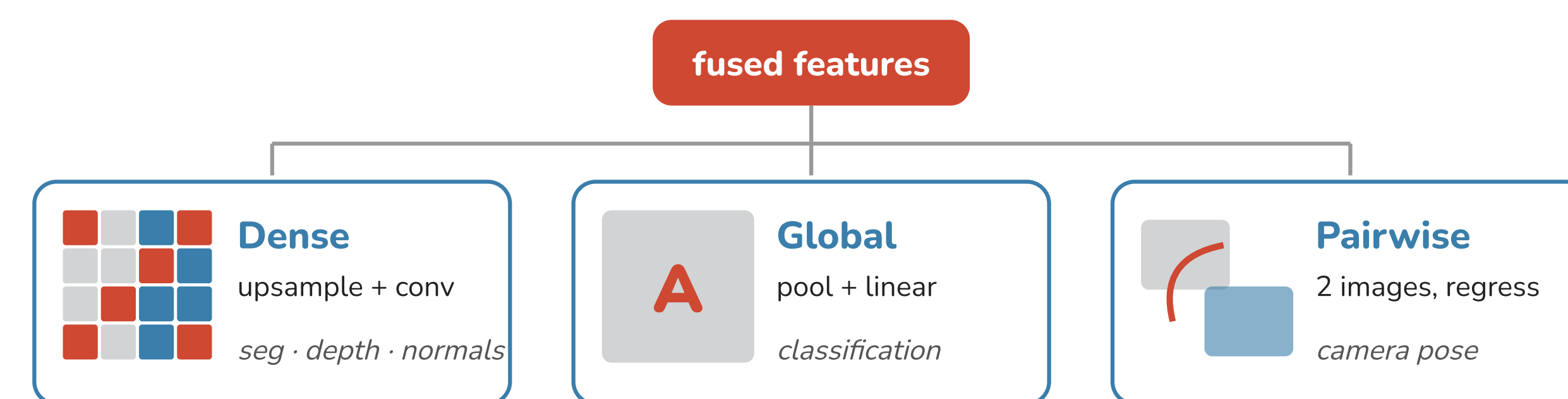


Fig. 4: The backbone stays frozen and all ViT blocks are used; only the fusion probe and the task head are trained.

Its attention follows layer utility

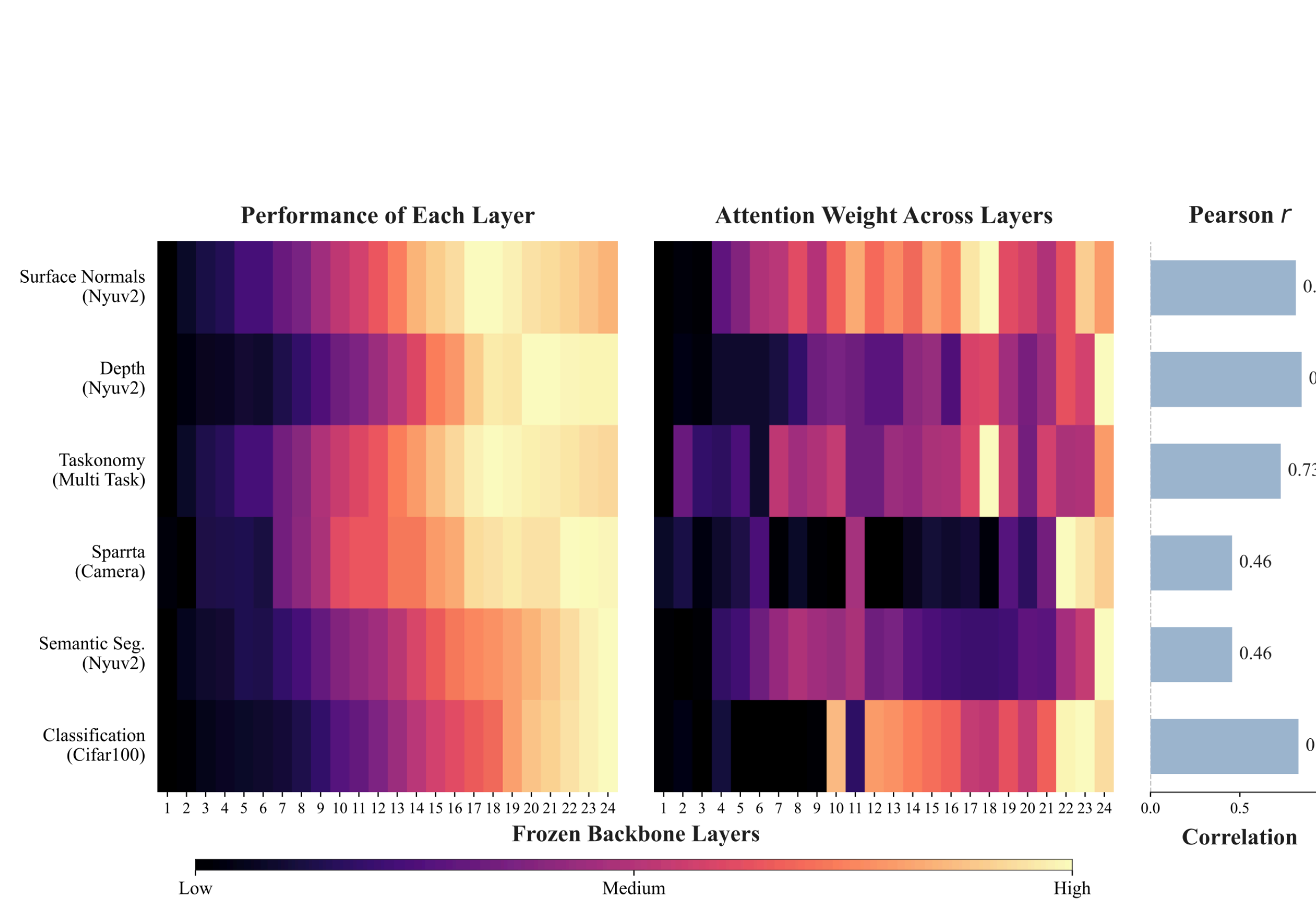


Fig. 5: Measured on an earlier AMDF variant (with per-layer refinement). AMDF's learned depth-attention (middle) tracks per-layer probe performance (left) on DINOv3-L/16. Alignment (Pearson r , right) is strongest for dense geometric tasks (depth $r \approx 0.85$, normals $r \approx 0.81$, Taskonomy $r \approx 0.73$) and shifts toward deep layers for semantic tasks (classification $r \approx 0.83$).

Results

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benchmarks where AMDF ranks best

16

expert queries, each picking its own depth

7×6

backbones × benchmarks

Mean rank per benchmark (1 = best, ★ = best in table)

	Normals	Depth	Segm.	Taskonomy	Pose	CUB-200
AMDF (ours)	1.00 ★	1.86	1.21 ★	1.29 ★	2.04	2.43
ConcatLayers	2.00	1.14 ★	2.43	1.71	1.32 ★	4.57
WeightedLayers	3.14	3.43	3.86	3.14	3.43	4.71
LastLayer	4.57	4.00	3.21	4.43	4.25	–
AvgLayer	4.29	4.57	4.29	4.43	3.96	6.29
EfficientProbe	–	–	–	–	–	1.14 ★
AttentiveFusionAP	–	–	–	–	–	3.00
AvgPool	–	–	–	–	–	5.86

Fig. 6: Each cell is a method's rank averaged over the 7 backbones of that table (NAVI: rotation and translation). AMDF is best on normals, segmentation and Taskonomy; ConcatLayers on depth and pose; EfficientProbe on CUB-200. Full tables: handout.

Accuracy vs. trainable parameters

	Mean rank ↓ (4 dense + pose)	Trainable params (M)
AMDF (ours)	1.48	1.79
ConcatLayers	1.72	6.99
WeightedLayers	3.40	0.70
LastLayer	4.09	0.70
AvgLayer	4.31	0.70

Fig. 7: Mean of the per-table ranks over 5 benchmarks (NYUv2 normals, depth, segmentation, Taskonomy, NAVI pose); params (probe + head, ViT-L) averaged over the same tasks. CUB-200 excluded (different methods).

Take-home

- Useful features are **spread across depth**, and the best depth depends on the task.
- AMDF has a better mean rank than LastLayer, AvgLayer and WeightedLayers in every table, and leads ConcatLayers on normals, segmentation and Taskonomy with **1/4 of its parameters**.
- On fine-grained classification, a dedicated last-layer probe (EfficientProbe) remains stronger.
- Study frozen VFMs as **full hierarchies**; no manual layer search needed.